

```
In [1]: import pandas as pd
import numpy as np
import xgboost as xgb
from sklearn.metrics import r2_score, root_mean_squared_error
from sklearn.model_selection import train_test_split
```

```
In [5]: Q_com = pd.read_excel("Q_Commerce_Data_rating_Count.xlsx")
Amazon = pd.read_excel("focus_products.xlsx")
Q_com
Amazon
```

Out[5]:

|    | product-name                                         | Pareto %<br>Products | Sum of absolute-<br>quantity-gms | Sum of<br>Net_Price |
|----|------------------------------------------------------|----------------------|----------------------------------|---------------------|
| 0  | FZYEZY Freeze Dried Apple Slices<br>for Kids and ... | 0.3962               | 140                              | 1000                |
| 1  | FZYEZY Freeze Dried Black Jamun<br>Fruit for Kids... | 0.1293               | 1740                             | 6120                |
| 2  | FZYEZY Freeze Dried Custard Apple<br>Fruit Powder... | 0.6846               | 600                              | 2320                |
| 3  | FzyEzy Freeze Dried Fruit Powder for<br>Gourmet      | 0.7511               | 300                              | 2874                |
| 4  | FzyEzy Freeze Dried Gajar Ka Halwa                   | 0.7909               | 600                              | 1250                |
| 5  | FzyEzy Freeze Dried Ginger<br>Vegetable Powder fo... | 0.7713               | 850                              | 2550                |
| 6  | FZYEZY Freeze Dried Mango Fruit<br>Powder for Kid... | 0.3424               | 1750                             | 4284                |
| 7  | FZYEZY Freeze Dried Pink Guava<br>Powder for Kids... | 0.5905               | 950                              | 3188                |
| 8  | FzyEzy Freeze-Dried Apple Powder -<br>100% Natura... | 0.4977               | 900                              | 4171                |
| 9  | FzyEzy Freeze-Dried Celery Powder -<br>100% Natur... | 0.2415               | 1362                             | 6150                |
| 10 | FzyEzy Freeze-Dried Jain Kadhi<br>Khichdi            | 0.7082               | 672                              | 2100                |
| 11 | FzyEzy Freeze-Dried Jain Paneer<br>Bhurji            | 0.5140               | 770                              | 2200                |
| 12 | FzyEzy Freeze-Dried Jain Paneer<br>Butter Masala     | 0.3785               | 750                              | 2100                |
| 13 | FzyEzy Freeze-Dried Kiwi Fruit<br>Powder - 100% N... | 0.7194               | 454                              | 3157                |
| 14 | FzyEzy Freeze-Dried Masala Kadhi<br>Khichdi          | 0.1754               | 720                              | 2250                |
| 15 | FzyEzy Freeze-Dried Palak Paneer                     | 0.6596               | 350                              | 1400                |
| 16 | FzyEzy Freeze-Dried Paneer Bhurji                    | 0.6723               | 420                              | 1200                |
| 17 | FzyEzy Freeze-Dried Paneer Gotala                    | 0.0545               | 700                              | 2200                |
| 18 | FzyEzy Freeze-Dried Spinach<br>Powder                | 0.7409               | 650                              | 2306                |
| 19 | FzyEzy Freeze-Dried Spinach<br>Powder - 100% Natu... | 0.7812               | 908                              | 2800                |

|    | product-name                                      | Pareto %<br>Products | Sum of absolute-<br>quantity-gms | Sum of<br>Net_Price |
|----|---------------------------------------------------|----------------------|----------------------------------|---------------------|
| 20 | FzyEzy Freeze-dried Strawberry Fruit Powder       | 0.5758               | 900                              | 4194                |
| 21 | FzyEzy Freeze-Dried Veg. Makhanwala               | 0.4814               | 600                              | 1800                |
| 22 | FzyEzy Healthy Dried Fruits Candy/Chips/Slices    | 0.4479               | 792                              | 4123                |
| 23 | FzyEzy Healthy Natural Premium Freeze Dried Ma... | 0.3042               | 3600                             | 5592                |
| 24 | FzyEzy Healthy Natural Premium Freeze Dried Pi... | 0.6324               | 908                              | 3902                |
| 25 | FzyEzy Healthy Natural Premium Freeze Dried Po... | 0.0800               | 7500                             | 7200                |
| 26 | FzyEzy Healthy Natural Premium Freeze Dried St... | 0.7303               | 600                              | 2640                |
| 27 | FzyEzy Healthy Natural Premium Freeze Dried Wh... | 0.6965               | 550                              | 2814                |
| 28 | FzyEzy Healthy Premium Freeze Dried Banana Fru... | 0.2198               | 2270                             | 6244                |
| 29 | FzyEzy Healthy Premium Freeze-dried Banana Fru... | 0.7997               | 750                              | 2463                |
| 30 | FzyEzy Healthy Premium Freeze-dried Beet Veget... | 0.1526               | 1150                             | 4089                |
| 31 | FzyEzy Healthy Premium Freeze-Dried Jackfruit ... | 0.3237               | 520                              | 3109                |
| 32 | FzyEzy Healthy Premium Freeze-Dried Kiwi Fruit... | 0.5297               | 340                              | 2028                |
| 33 | FZYEZY Natural Freeze Dried Apple Powder for K... | 0.7612               | 550                              | 2420                |
| 34 | FZYEZY Natural Freeze Dried Banana Powder for ... | 0.0281               | 3250                             | 6705                |
| 35 | FZYEZY Natural Freeze Dried Black Jamun Chunks... | 0.1053               | 1300                             | 4299                |
| 36 | FZYEZY Natural Freeze Dried Black Jamun for Ki... | 0.5453               | 1200                             | 3060                |
| 37 | FZYEZY Natural Freeze Dried Chikoo Fruit for K... | 0.1976               | 1660                             | 4670                |
| 38 | FZYEZY Natural Freeze Dried Chikoo Fruit Powde... | 0.2844               | 1500                             | 4500                |

|    | product-name                                      | Pareto % Products | Sum of absolute-quantity-gms | Sum of Net_Price |
|----|---------------------------------------------------|-------------------|------------------------------|------------------|
| 39 | FZYEZY Natural Freeze Dried Mango Powder for K... | 0.4307            | 800                          | 4850             |
| 40 | FZYEZY Natural Freeze Dried Pineapple Slices f... | 0.3606            | 680                          | 2740             |
| 41 | FzyEzy Natural Freeze Dried Pink Guava Powder ... | 0.6185            | 800                          | 3489             |
| 42 | FzyEzy Premium Moringa Vegetable Powder           | 0.6461            | 3859                         | 3888             |
| 43 | FZYEZY Pure & Natural Premium Mango Fruit Powd... | 0.6046            | 400                          | 1998             |
| 44 | Hamps Heamwalt - Whey Protein, Vitamins, Miner... | 0.2632            | 9800                         | 5367             |
| 45 | Oxynxt Freeze-dried Beet Powder Sachets-10 Count  | 0.4136            | 560                          | 1032             |
| 46 | Oxynxt Freeze-dried Daily Green Powder Sachets... | 0.4648            | 400                          | 2392             |
| 47 | Oxynxt Freeze-dried Prickly Pear Powder Sachet... | 0.5609            | 770                          | 2132             |

```
In [42]: Q_com.columns = Q_com.columns.str.strip()
Amazon.columns = Amazon.columns.str.strip()
Q_com.columns
Amazon.columns
```

```
Out[42]: Index(['product-name', 'Pareto % Products', 'Sum of absolute-quantity-gms',
               'Sum of Net_Price'],
              dtype='object')
```

```
In [43]: Q_com.columns
```

```
Out[43]: Index(['Brand', 'Quantity', 'MRP', 'Price', 'Price_100_g', 'Discount',
               'Rating', 'Rating_Count'],
              dtype='object')
```

```
In [ ]:
```

```
In [9]: Features = ['Quantity', 'Price']
Target = 'Rating_Count'
```

```
In [24]: Q_com_features = Q_com[Features].copy()
Q_com_target = Q_com[Target].copy()
```

```
In [30]: Q_com_features
Q_com_target
```

```
Out[30]: 0      846
         1     1300
         2     4000
         3     1900
         4     1300
         ...
        70     121
        71     192
        72     242
        73     202
        74     195
        Name: Rating_Count, Length: 75, dtype: int64
```

```
In [32]: Q_com_features.fillna(Q_com_features.median(),inplace=True)
         Q_com_target.fillna(Q_com_target.median(),inplace=True)
```

```
In [34]: Q_com_features_train , Q_com_features_test , Q_com_target_train ,Q_com_target_test
         model = xgb.XGBRegressor(
             objective= "reg:squarederror",
             n_estimator= 100,
             learning_rate= 0.1,
             max_depth= 5,
             random_state= 42
         )
         model.fit(Q_com_features_train,Q_com_target_train)
         Q_com_pred = model.predict(Q_com_features_test)
```

C:\Users\Ritwik Singh\AppData\Roaming\Python\Python313\site-packages\xgboost\training.py:200: UserWarning: [19:19:17] WARNING: C:\actions-runner\work\xgboost\xgboost\src\learner.cc:782:  
Parameters: { "n\_estimator" } are not used.

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [37]: rmse = (root_mean_squared_error(Q_com_target_test,Q_com_pred))
         r2 = (r2_score(Q_com_target_test,Q_com_pred))*100
```

```
In [40]: print(f"RMSE (Root mean Squared Error) = {rmse:.2f}")
         print(f"R^2 error = {r2:.2f} %")
```

RMSE (Root mean Squared Error) = 2594.80  
R^2 error = 82.04 %

```
In [44]: Amazon.rename(columns={'Sum of absolute-quantity-gms':'Quantity',
                               'Sum of Net_Price': 'Price'
                               }, inplace=True)
```

```
In [48]: Amazon_features = Amazon[Features]
         Amazon_features
```

Out[48]:

|           | Quantity | Price |
|-----------|----------|-------|
| <b>0</b>  | 140      | 1000  |
| <b>1</b>  | 1740     | 6120  |
| <b>2</b>  | 600      | 2320  |
| <b>3</b>  | 300      | 2874  |
| <b>4</b>  | 600      | 1250  |
| <b>5</b>  | 850      | 2550  |
| <b>6</b>  | 1750     | 4284  |
| <b>7</b>  | 950      | 3188  |
| <b>8</b>  | 900      | 4171  |
| <b>9</b>  | 1362     | 6150  |
| <b>10</b> | 672      | 2100  |
| <b>11</b> | 770      | 2200  |
| <b>12</b> | 750      | 2100  |
| <b>13</b> | 454      | 3157  |
| <b>14</b> | 720      | 2250  |
| <b>15</b> | 350      | 1400  |
| <b>16</b> | 420      | 1200  |
| <b>17</b> | 700      | 2200  |
| <b>18</b> | 650      | 2306  |
| <b>19</b> | 908      | 2800  |
| <b>20</b> | 900      | 4194  |
| <b>21</b> | 600      | 1800  |
| <b>22</b> | 792      | 4123  |
| <b>23</b> | 3600     | 5592  |
| <b>24</b> | 908      | 3902  |
| <b>25</b> | 7500     | 7200  |
| <b>26</b> | 600      | 2640  |
| <b>27</b> | 550      | 2814  |
| <b>28</b> | 2270     | 6244  |
| <b>29</b> | 750      | 2463  |

|           | Quantity | Price |
|-----------|----------|-------|
| <b>30</b> | 1150     | 4089  |
| <b>31</b> | 520      | 3109  |
| <b>32</b> | 340      | 2028  |
| <b>33</b> | 550      | 2420  |
| <b>34</b> | 3250     | 6705  |
| <b>35</b> | 1300     | 4299  |
| <b>36</b> | 1200     | 3060  |
| <b>37</b> | 1660     | 4670  |
| <b>38</b> | 1500     | 4500  |
| <b>39</b> | 800      | 4850  |
| <b>40</b> | 680      | 2740  |
| <b>41</b> | 800      | 3489  |
| <b>42</b> | 3859     | 3888  |
| <b>43</b> | 400      | 1998  |
| <b>44</b> | 9800     | 5367  |
| <b>45</b> | 560      | 1032  |
| <b>46</b> | 400      | 2392  |
| <b>47</b> | 770      | 2132  |

```
In [51]: predicted_ratings = model.predict(Amazon_features)
predicted_ratings = np.maximum(0,predicted_ratings)
predicted_ratings = np.round(predicted_ratings).astype(int)
```

```
In [52]: Amazon["Projected_customer_ratings"] = predicted_ratings
```

```
In [55]: Amazon.to_excel("Amazon_Predicted_Customer_count.xlsx")
```